Please find a few remarks to help you get started:

Our sales process starts at “Opportunity Accepted” and goes until “Deploy”, where the opportunity is considered won. “Closed Lost” indicates opportunities that were pursued but not won. Qualified Out means the opportunity was presented to our sales Manager but was not accepted. Lead Source indicates where the opportunity originated, while Campaign reflects the marketing initiative that generated it. “Meeting Set”, “Meeting Rescheduled”, and “Meeting Finished” are opportunity stages prior to “Opportunity Accepted”. These are primarily used by our SDR and Alliance teams (and occasionally others) to pass potential opportunities to Sales.

Based on this data, please share the top three key insights that you believe the company should be aware of. Kindly focus your analysis specifically on opportunities of type “New Business” and “Upsells.”

In [263…

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import openpyxl

import warnings
warnings.filterwarnings('ignore')
```

In [263…

```
# Reading the sales opportunity data from an Excel file

df = pd.read_excel('Test Data For Customer Success.xlsx')
```

Basic details of the provided data

In [263…

```
df.head()
```

Out[263…

	Opportunity Owner	Account Name	Opportunity Name	Stage	Fiscal Period	Annual Contract Value Currency	Annual Contract Value	Probability (%)
0	Sales Manager 11	Customer 1	Opportunity 1	Closed Lost	Q3-2024	EUR	80000.0	0
1	Sales Manager 8	Customer 2	Opportunity 2	Qualified Out	Q3-2024	USD	200000.0	0
2	Sales Manager 6	Customer 3	Opportunity 3	Closed Lost	Q4-2024	USD	550000.0	0
3	Sales Manager 11	Customer 4	Opportunity 4	Closed Lost	Q2-2024	EUR	250000.0	0
4	Sales Manager 3	Customer 5	Opportunity 5	Qualified Out	Q4-2024	USD	269500.0	0

In [263…

```
df.describe().round()
```

Out[263…

	Annual Contract Value	Probability (%)	Age	Estimated Close Date	Created Date
count	400.0	555.0	555.0	555	555
mean	134180.0	16.0	291.0	2024-08-05 11:58:42.162162176	2023-11-26 07:49:37.297297408
min	0.0	0.0	0.0	2024-01-01 00:00:00	2021-09-14 00:00:00
25%	19750.0	0.0	230.0	2024-05-30 12:00:00	2023-08-08 00:00:00
50%	100000.0	0.0	297.0	2024-08-15 00:00:00	2024-01-11 00:00:00
75%	180000.0	25.0	320.0	2024-10-31 00:00:00	2024-03-18 00:00:00
max	1350000.0	100.0	1117.0	2024-12-31 00:00:00	2025-01-08 00:00:00
std	165218.0	31.0	160.0	NaN	NaN

In [263…

```
df.columns
```

Out[263…

```
Index(['Opportunity Owner', 'Account Name', 'Opportunity Name', 'Stage',
       'Fiscal Period', 'Annual Contract Value Currency',
       'Annual Contract Value', 'Probability (%)', 'Age',
       'Estimated Close Date', 'Created Date', 'Lead Source',
       'Primary Campaign Source', 'Opportunity Type'],
      dtype='object')
```

In [263…

```
#Data types of each column

df.dtypes
```

Out[263…

```
Opportunity Owner                        object
Account Name                             object
Opportunity Name                         object
Stage                                    object
Fiscal Period                            object
Annual Contract Value Currency           object
Annual Contract Value                   float64
Probability (%)                           int64
Age                                       int64
Estimated Close Date              datetime64[ns]
Created Date                      datetime64[ns]
Lead Source                              object
Primary Campaign Source                  object
Opportunity Type                         object
dtype: object
```

In [263…

```
#Shape of the DataFrame
df.shape
```

Out[263…

```
(555, 14)
```

In [264…

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 555 entries, 0 to 554
Data columns (total 14 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Opportunity Owner               555 non-null    object
 1   Account Name                    555 non-null    object
 2   Opportunity Name                555 non-null    object
 3   Stage                           555 non-null    object
 4   Fiscal Period                   555 non-null    object
 5   Annual Contract Value Currency  400 non-null    object
 6   Annual Contract Value           400 non-null    float64
 7   Probability (%)                 555 non-null    int64
 8   Age                             555 non-null    int64
 9   Estimated Close Date            555 non-null    datetime64[ns]
 10  Created Date                    555 non-null    datetime64[ns]
 11  Lead Source                     434 non-null    object
 12  Primary Campaign Source         167 non-null    object
 13  Opportunity Type                548 non-null    object
dtypes: datetime64[ns](2), float64(1), int64(2), object(9)
memory usage: 60.8+ KB
```

- Some of the columns has null values

PRE-PROCESSING OF THE DATA

Checking Duplicate Records

In [264…

```
# Removig duplicates
df.duplicated().sum()

# Zero duplicate rows found
```

Out[264…

```
np.int64(0)
```

Checking inconsistency with 'Opportunity Name' column

In [264…

```
# Checking whether there is any inconsistency in the opportunity name column
# Opportunity need to be unique in order to proceed for the analysis

df['Opportunity_Name_Pre'] = df['Opportunity Name'].apply(lambda x: x.strip().upper
Opporunity_values = df['Opportunity_Name_Pre'].value_counts().index.str.strip()

# With this check we indentified that there are no inconsistencies in the opportuni
```

In [264…

```
Opporunity_values = [int(i) for i in Opporunity_values]
Actual_oppotunity = [i for i in range(1, max(Opporunity_values)+1, 1)]

# Missing opportunity details [SOFT CHECK]
Missing_opportunity_data = set(Actual_oppotunity) - set(Opporunity_values)

print('Data missing for the following opportunity numbers:')
print(sorted(Missing_opportunity_data))
print('Does not impact the analysis as these opportunity numbers are not present in
```

```
Data missing for the following opportunity numbers:
[13, 25, 43, 57, 58, 60, 61, 62, 71, 77, 88, 91, 98, 99, 100, 110, 133, 137, 138, 14
3, 147, 150, 175, 180, 185, 199, 201, 243, 257, 282, 285, 316, 329, 573]
Does not impact the analysis as these opportunity numbers are not present in the dat
aset.
```

In [264…

```
# --- IGNORE ---
```

Dropping 'Account Name', 'Opportunity_Name_Pre', 'Opportunity Name'

In [264…

```
df['Account Name'].value_counts().sum()

# All are unique account names so can be dropped from the analysis
```

Out[264…

```
np.int64(555)
```

In [264…

```
df.drop(axis=1, columns = ['Account Name', 'Opportunity_Name_Pre', 'Opportunity Nam
```

In [264…

```
df
```

Out[264…

	Opportunity Owner	Stage	Fiscal Period	Annual Contract Value Currency	Annual Contract Value	Probability (%)	Age	Estimated Close Date	Crea D
0	Sales Manager 11	Closed Lost	Q3-2024	EUR	80000.0	0	1098	2024-09-15	2 0
1	Sales Manager 8	Qualified Out	Q3-2024	USD	200000.0	0	969	2024-07-12	2 1
2	Sales Manager 6	Closed Lost	Q4-2024	USD	550000.0	0	1117	2024-12-20	2 1
3	Sales Manager 11	Closed Lost	Q2-2024	EUR	250000.0	0	785	2024-06-27	2 0
4	Sales Manager 3	Qualified Out	Q4-2024	USD	269500.0	0	870	2024-10-25	2 0
...	...	...	...	...	...	...	...	...	
550	Sales Manager 14	Qualified Out	Q3-2024	NaN	NaN	0	73	2024-08-12	2 0
551	Sales Manager 12	Deploy	Q4-2024	USD	3260.0	100	0	2024-10-02	2 1
552	Sales Manager 16	Deploy	Q4-2024	USD	8000.0	100	12	2024-12-09	2 1
553	Sales Manager 12	Deploy	Q4-2024	USD	6254.0	100	1	2024-12-31	2 1
554	Sales Manager 15	Deploy	Q4-2024	USD	1910.0	100	0	2024-12-31	2 0

555 rows × 12 columns

Null value handling

In [264…

```
# Working on Null values

# Count of null values in each column

Null_values_count = df.isnull().sum()
Null_values_count[Null_values_count > 0]
```

Out[264…

```
Annual Contract Value Currency    155
Annual Contract Value             155
Lead Source                       121
Primary Campaign Source           388
Opportunity Type                    7
dtype: int64
```

In [264…

```
# Percentage of null values in each column

Null_values_per = round((df.isnull().sum()/df.shape[0] )* 100, 2)
Null_values_per[Null_values_per > 0]
```

Out[264…

```
Annual Contract Value Currency    27.93
Annual Contract Value             27.93
Lead Source                       21.80
Primary Campaign Source           69.91
Opportunity Type                   1.26
dtype: float64
```

Dropping of Primary Campaign Source

In [265…

```
# Values in Primary Campaign Source column are missing for around 69.91% of the dat
df.drop(axis=1, columns = ['Primary Campaign Source'], inplace=True)
```

Handling of Null values in 'Opportunity Type' column

In [265…

```
# Values in Opportunity type column are missing for around 1% of the data. Hence re
df['Opportunity Type'].fillna(df['Opportunity Type'].mode()[0], inplace=True)
```

Dropping Estimated Close Date' & 'Created Date'

In [265…

```
# As Age is the column which is difference between Estimated Close Date & Created D
df.drop(axis = 1, columns = ['Estimated Close Date', 'Created Date'], inplace=True)
```

In [265…

```
df.head()
```

Out[265…

	Opportunity Owner	Stage	Fiscal Period	Annual Contract Value Currency	Annual Contract Value	Probability (%)	Age	Lead Source	Oppor
0	Sales Manager 11	Closed Lost	Q3-2024	EUR	80000.0	0	1098	DHL Referral	Bu
1	Sales Manager 8	Qualified Out	Q3-2024	USD	200000.0	0	969	SDR Team	Bu
2	Sales Manager 6	Closed Lost	Q4-2024	USD	550000.0	0	1117	Outside Sales	Bu
3	Sales Manager 11	Closed Lost	Q2-2024	EUR	250000.0	0	785	Marketing	Bu
4	Sales Manager 3	Qualified Out	Q4-2024	USD	269500.0	0	870	Customer Success	U

In [265…

```
df.isnull().sum()
```

Out[265…

```
Opportunity Owner                   0
Stage                               0
Fiscal Period                       0
Annual Contract Value Currency    155
Annual Contract Value             155
Probability (%)                     0
Age                                 0
Lead Source                       121
Opportunity Type                    0
dtype: int64
```

Handling of 'Annual Contract Value Currency' & 'Annual Contract Value' for ('Closed Lost', 'Qualified Out')

In [265…

```
# When stage of the opportunity is Closed Lost or Qualified Out, will not have any
# Replacing the null values in these columns with 0

# For the 'No Currency' fill
df.loc[df['Stage'].isin(['Closed Lost', 'Qualified Out']), 'Annual Contract Value C
df.loc[df['Stage'].isin(['Closed Lost', 'Qualified Out']), 'Annual Contract Value']
```

Handling of 'Lead Source'

In [265…

```
# As the Lead Source is ~20% missing, we will replace the missing values with Mode
df['Lead Source'].fillna(df['Lead Source'].mode()[0], inplace=True)
```

In [265…

```
df.isnull().sum()
```

Out[265…

```
Opportunity Owner                   0
Stage                               0
Fiscal Period                       0
Annual Contract Value Currency     17
Annual Contract Value              17
Probability (%)                     0
Age                                 0
Lead Source                         0
Opportunity Type                    0
dtype: int64
```

In [265…

```
df[df['Annual Contract Value Currency'].isnull()].loc[:, 'Stage']
```

Out[265…

```
95     Confirm Value
98     Confirm Value
103    Confirm Value
104    Confirm Value
105    Confirm Value
107    Confirm Value
108    Confirm Value
109    Confirm Value
110    Confirm Value
112    Confirm Value
113           Deploy
114           Deploy
115           Deploy
116           Deploy
117           Deploy
119          Nurture
121          Nurture
Name: Stage, dtype: object
```

Excluded Stage = 'Confirm Value', 'Nurture' from the analysis as they are INPROGRESS - Can be considered into analysis post acceptance only

In [265…

```
# Lets exclude stage = 'Confirm Value' &  'Nurture' from the analysis as these are
df = df[~df['Stage'].isin(['Confirm Value', 'Nurture'])]
```

In [266…

```
df.isnull().sum()
```

Out[266…

```
Opportunity Owner                 0
Stage                             0
Fiscal Period                     0
Annual Contract Value Currency    5
Annual Contract Value             5
Probability (%)                   0
Age                               0
Lead Source                       0
Opportunity Type                  0
dtype: int64
```

Handling of 'Annual Contract Value Currency' & 'Annual Contract Value' for ('Deploy')

In [266…

```
# As the Stage = 'Deploy' is important lets fill the missing values in Annual Contr
mode_deploy = df[df['Stage'] == 'Deploy']['Annual Contract Value Currency'].mode()[
median_deploy = df[(df['Stage'] == 'Deploy') & (df['Annual Contract Value Currency'


df.loc[df['Stage'].isin(['Deploy']), 'Annual Contract Value Currency'] = df.loc[df[
df.loc[df['Stage'].isin(['Deploy']), 'Annual Contract Value'] = df.loc[df['Stage'].
```

In [266…

```
df.isnull().sum()
```

Out[266…

```
Opportunity Owner                 0
Stage                             0
Fiscal Period                     0
Annual Contract Value Currency    0
Annual Contract Value             0
Probability (%)                   0
Age                               0
Lead Source                       0
Opportunity Type                  0
dtype: int64
```

In [266…

```
df.head()
```

Out[266…

	Opportunity Owner	Stage	Fiscal Period	Annual Contract Value Currency	Annual Contract Value	Probability (%)	Age	Lead Source	Oppor
0	Sales Manager 11	Closed Lost	Q3-2024	EUR	80000.0	0	1098	DHL Referral	Bu
1	Sales Manager 8	Qualified Out	Q3-2024	USD	200000.0	0	969	SDR Team	Bu
2	Sales Manager 6	Closed Lost	Q4-2024	USD	550000.0	0	1117	Outside Sales	Bu
3	Sales Manager 11	Closed Lost	Q2-2024	EUR	250000.0	0	785	Marketing	Bu
4	Sales Manager 3	Qualified Out	Q4-2024	USD	269500.0	0	870	Customer Success	U

In [266…

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 423 entries, 0 to 554
Data columns (total 9 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Opportunity Owner               423 non-null    object
 1   Stage                           423 non-null    object
 2   Fiscal Period                   423 non-null    object
 3   Annual Contract Value Currency  423 non-null    object
 4   Annual Contract Value           423 non-null    float64
 5   Probability (%)                 423 non-null    int64
 6   Age                             423 non-null    int64
 7   Lead Source                     423 non-null    object
 8   Opportunity Type                423 non-null    object
dtypes: float64(1), int64(2), object(6)
memory usage: 33.0+ KB
```

Exploratory data analysis

In [266…

```
dataset = df[df['Opportunity Type'].isin(['New Business', 'Up-Sell'])]
```

In [266…

```
dataset.shape
```

Out[266…

```
(400, 9)
```

In [266…

```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 400 entries, 0 to 554
Data columns (total 9 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Opportunity Owner               400 non-null    object
 1   Stage                           400 non-null    object
 2   Fiscal Period                   400 non-null    object
 3   Annual Contract Value Currency  400 non-null    object
 4   Annual Contract Value           400 non-null    float64
 5   Probability (%)                 400 non-null    int64
 6   Age                             400 non-null    int64
 7   Lead Source                     400 non-null    object
 8   Opportunity Type                400 non-null    object
dtypes: float64(1), int64(2), object(6)
memory usage: 31.2+ KB
```

In [266…

```
dataset.head()
```

Out[266…

	Opportunity Owner	Stage	Fiscal Period	Annual Contract Value Currency	Annual Contract Value	Probability (%)	Age	Lead Source	Oppor
0	Sales Manager 11	Closed Lost	Q3-2024	EUR	80000.0	0	1098	DHL Referral	Bu
1	Sales Manager 8	Qualified Out	Q3-2024	USD	200000.0	0	969	SDR Team	Bu
2	Sales Manager 6	Closed Lost	Q4-2024	USD	550000.0	0	1117	Outside Sales	Bu
3	Sales Manager 11	Closed Lost	Q2-2024	EUR	250000.0	0	785	Marketing	Bu
4	Sales Manager 3	Qualified Out	Q4-2024	USD	269500.0	0	870	Customer Success	U

Opportunity Type Vs Stage

In [266…

```
dataset['Opportunity Type'].value_counts()
```

Out[266…

```
Opportunity Type
New Business    327
Up-Sell          73
Name: count, dtype: int64
```

In [267…

```
dataset.describe().round()
```

Out[267…

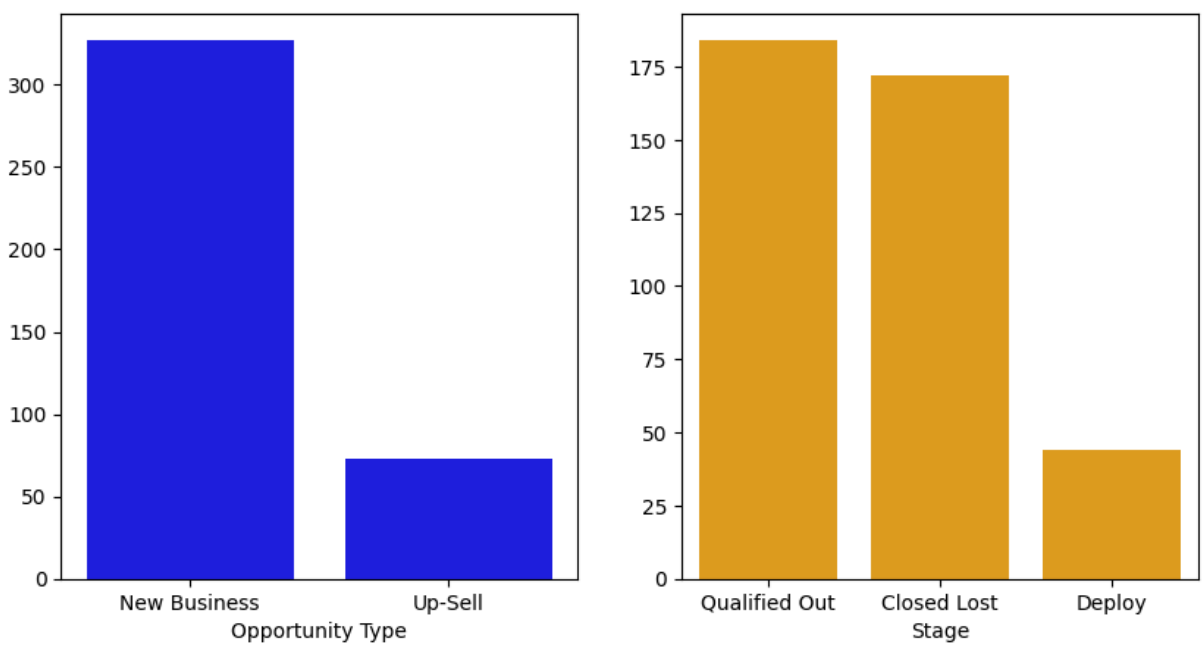
	Annual Contract Value	Probability (%)	Age
count	400.0	400.0	400.0
mean	111654.0	11.0	296.0
std	156787.0	31.0	179.0
min	0.0	0.0	0.0
25%	0.0	0.0	189.0
50%	79500.0	0.0	275.0
75%	150000.0	0.0	376.0
max	1350000.0	100.0	1117.0

In [267…

```
plt.figure(figsize= (10,5))

plt.subplot(1, 2, 1)
sns.barplot(x=dataset['Opportunity Type'].value_counts().index, y=dataset['Opportun

plt.subplot(1, 2, 2)
sns.barplot(x=dataset['Stage'].value_counts().index, y=dataset['Stage'].value_count
plt.show()
```



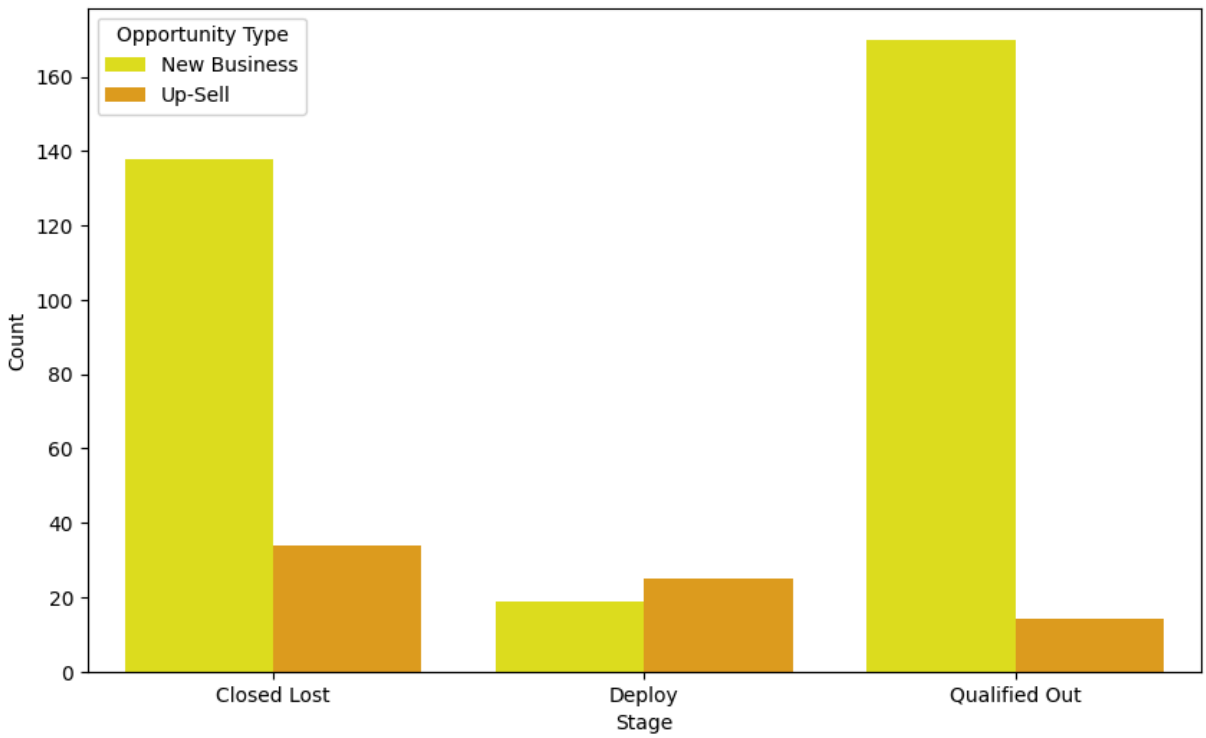
In [267…

```
# Create a summary table that actually has a column named 'Count'
summary = dataset.groupby(['Stage', 'Opportunity Type']).size().reset_index(name='C
```

```
plt.figure(figsize=(10, 6))

sns.barplot(
    data=summary[summary['Opportunity Type'].isin(['New Business', 'Up-Sell'])],
    x='Stage',
    y='Count',
    hue='Opportunity Type',
    palette={'New Business': 'Yellow', 'Up-Sell': 'Orange'}
)

plt.show()
```



In [267…

```
Win_df  = dataset[['Stage', 'Opportunity Type']].pivot_table(index='Stage', columns

Win_df
```

Out[267…

Opportunity Type	New Business	Up-Sell
Stage		
Closed Lost	138	34
Deploy	19	25
Qualified Out	170	14

In [267…

```
Win_df['Total'] = Win_df.sum(axis=1)

Win_df
```

Out[267...

Opportunity Type Stage	New Business	Up-Sell	Total
Closed Lost	138	34	172
Deploy	19	25	44
Qualified Out	170	14	184

Insight 1 - Volume vs wins

Total opportunities

- New Business: 138+19+170=327
- Upsell: 34+25+14=73.

Wins (Deploy)

- New Business: 19
- Upsell: 25.

So, New Business has far more volume, but fewer wins in absolute terms than Upsell, despite having >4× as many opportunities.

Insight 2 - Win-rate by type

This clearly shows:

The New Business funnel is very leaky: most deals end as Closed Lost or Qualified Out.

The Upsell motion is much healthier, converting roughly one-third of its opportunities to Deploy.

Conclusion :

- New Business win-rate ≈ 19/327≈6% 19/327≈6%.
- Upsell win-rate ≈ 25/73≈34%.
- New Business: high volume, very low win-rate, majority in Closed Lost + Qualified Out.
- Upsell: lower volume but higher proportion of Deploy, hence much better pipeline quality.

Opportunity Type Vs Probability

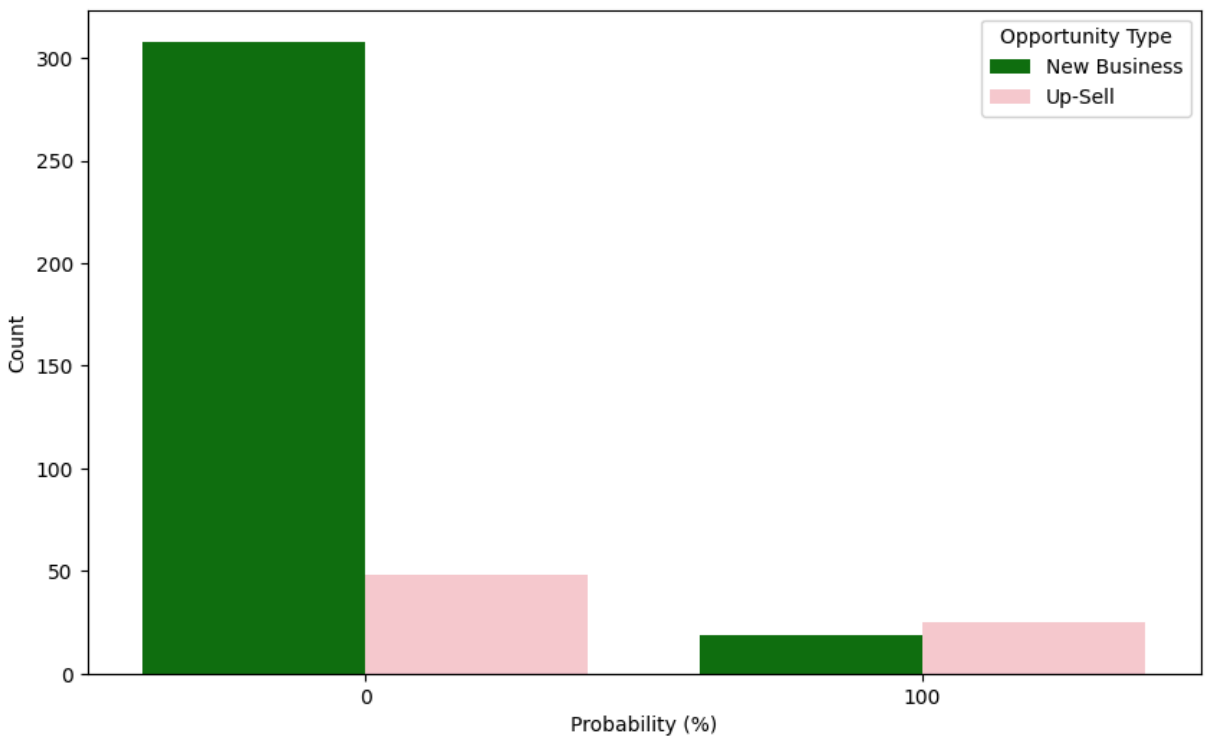
In [267…

```
# Create a summary table that actually has a column named 'Count'
summary = dataset.groupby(['Probability (%)', 'Opportunity Type']).size().reset_ind


plt.figure(figsize=(10, 6))

sns.barplot(
    data=summary[summary['Opportunity Type'].isin(['New Business', 'Up-Sell'])],
    x='Probability (%)',
    y='Count',
    hue='Opportunity Type',
    palette={'New Business': 'Green', 'Up-Sell': 'Pink'}
)

plt.show()
```



In [267…

```
summary.groupby(['Probability (%)', 'Opportunity Type']).sum()
```

Out[267…

Probability (%)	Opportunity Type	Count
0	**New Business**	308
	Up-Sell	48
100	**New Business**	19
	Up-Sell	25

Insight 3

- Almost the entire New Business pipeline and a large part of Upsell sit at 0% probability, indicating a lot of dead or non-winnable deals in the funnel.

Opportunity Owner Vs Opportunity Owner

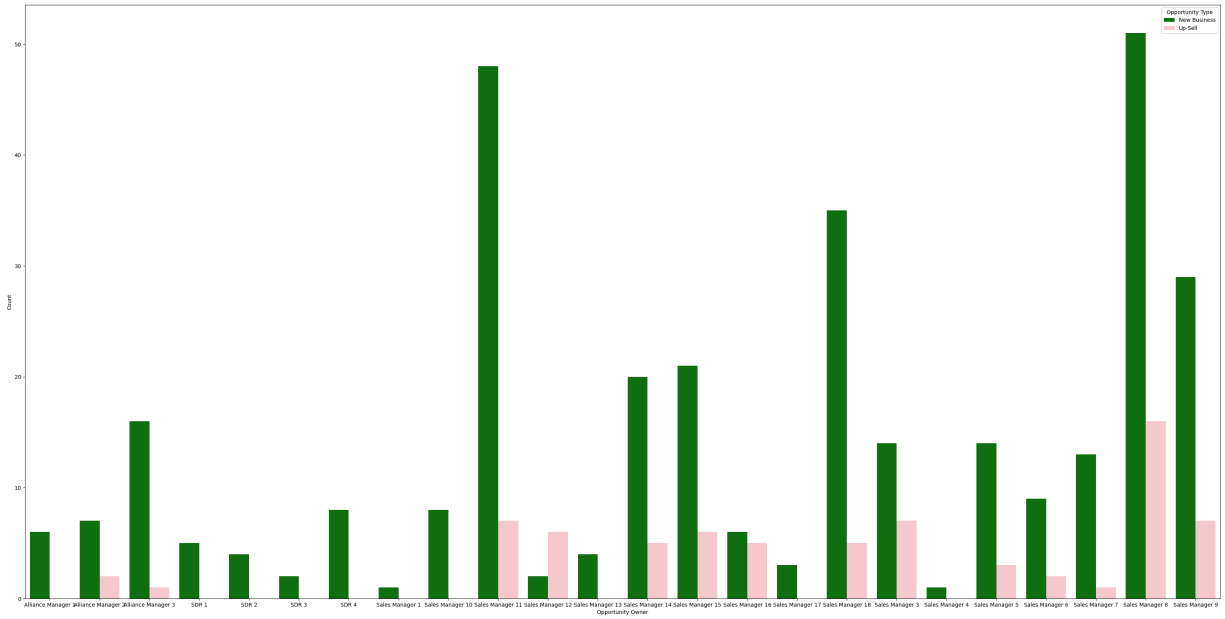
In [267…

```
# Create a summary table that actually has a column named 'Count'
summary = dataset.groupby(['Opportunity Owner', 'Opportunity Type']).size().reset_i

plt.figure(figsize=(40, 20))

sns.barplot(
    data=summary[summary['Opportunity Type'].isin(['New Business', 'Up-Sell'])],
    x='Opportunity Owner',
    y='Count',
    hue='Opportunity Type',
    palette={'New Business': 'Green', 'Up-Sell': 'Pink'}
)

plt.show()
```



In [267…

```
Owner_dataset = dataset[['Opportunity Owner', 'Opportunity Type', 'Stage']].pivot_t
```

In [267…

```
Owner_dataset['Win Rate'] = round(Owner_dataset['Deploy'] / (Owner_dataset['Closed
```

In [268…

```
Owner_dataset[Owner_dataset['Win Rate'] > 0].sort_values(by=['Opportunity Type', 'W
```

Out[268...

Stage	Opportunity Type	Opportunity Owner	Closed Lost	Deploy	Qualified Out	Win Rate
20	New Business	Sales Manager 6	8	1	0	11.11
12	New Business	Sales Manager 14	3	2	15	10.00
22	New Business	Sales Manager 8	33	5	13	9.80
9	New Business	Sales Manager 11	28	4	16	8.33
21	New Business	Sales Manager 7	0	1	12	7.69
23	New Business	Sales Manager 9	6	2	21	6.90
2	New Business	Alliance Manager 3	11	1	4	6.25
16	New Business	Sales Manager 18	15	2	18	5.71
13	New Business	Sales Manager 15	8	1	12	4.76
30	Up-Sell	Sales Manager 16	2	3	0	60.00
31	Up-Sell	Sales Manager 18	2	3	0	60.00
37	Up-Sell	Sales Manager 9	3	4	0	57.14
24	Up-Sell	Alliance Manager 2	1	1	0	50.00
29	Up-Sell	Sales Manager 15	2	3	1	50.00
26	Up-Sell	Sales Manager 11	4	3	0	42.86
36	Up-Sell	Sales Manager 8	11	5	0	31.25
28	Up-Sell	Sales Manager 14	2	1	2	20.00
27	Up-Sell	Sales Manager 12	4	1	1	16.67
32	Up-Sell	Sales Manager 3	0	1	6	14.29

Insight 4

- Best Opportunity Owner - Up-Sell : Sales Manager 16, Sales Manager 18
- Best Opportunity Owner - New Business : Sales Manager 6
- Upsell owners convert far better (often 50–60%), showing a much stronger upsell motion.
- Best New Business owners still have low win rates (about 5–11%), so New Business conversion is weak overall.

Opportunity Owner Vs Currency

In [268…

```
Currency_dataset = dataset[['Annual Contract Value Currency', 'Opportunity Type', '

Currency_dataset
```

Out[268…

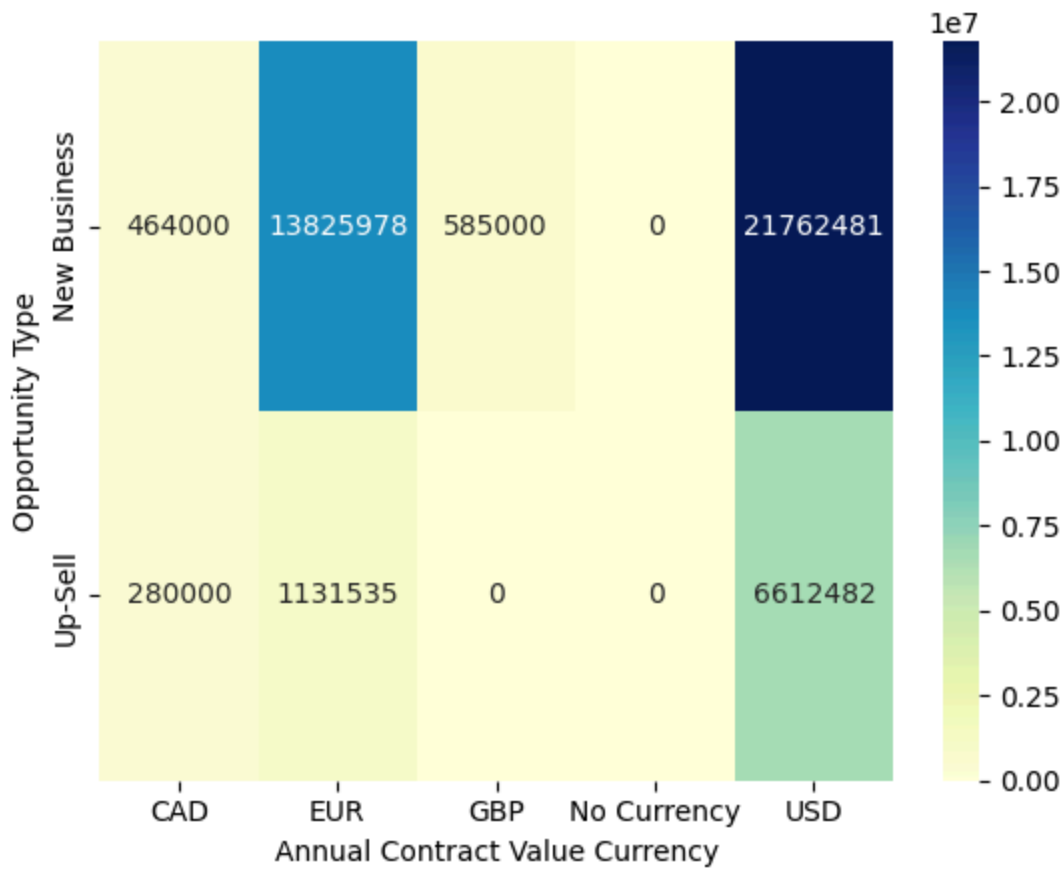
Annual Contract Value Currency Opportunity Type	CAD	EUR	GBP	No Currency	USD
New Business	464000.0	13825977.54	585000.0	0.0	21762481.00
Up-Sell	280000.0	1131535.00	0.0	0.0	6612481.62

In [268…

```
sns.heatmap(Currency_dataset, annot=True, fmt='.0f', cmap='YlGnBu')
```

Out[268…

```
<Axes: xlabel='Annual Contract Value Currency', ylabel='Opportunity Type'>
```



Insight 5:

- New Business is heavily concentrated in USD and EUR (about 21.76M USD and 13.83M EUR), with much smaller totals in CAD and GBP.

- Upsell revenue is also mainly USD and EUR, but on a smaller scale (about 6.61M USD and 1.13M EUR).
- This means overall revenue depends strongly on a few currencies/regions, so any slowdown or pricing issue there would hit both New Business and Upsell, which is exactly the concentration risk you are describing.

In [ ]:

In [ ]:

In [ ]: